

Buck-on Set-up Instructions

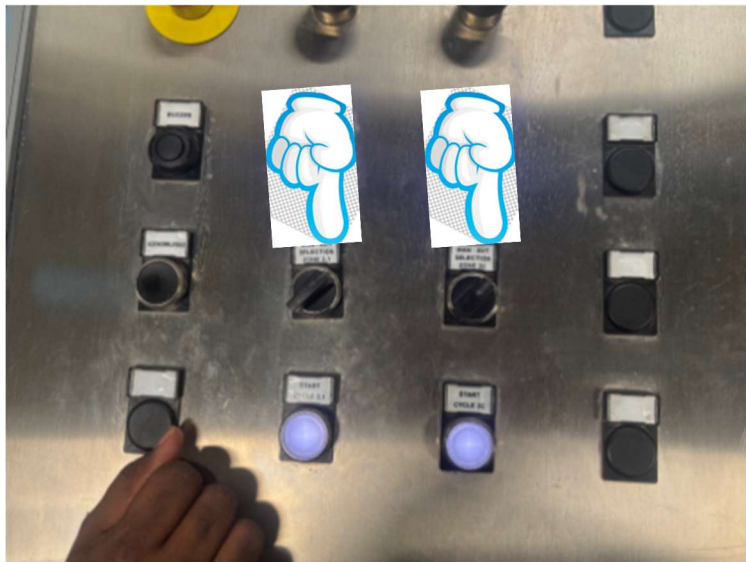
Pre-Setup Checklist

Before beginning the setup process, ensure the last pipe has completely exited the Buck-on area.

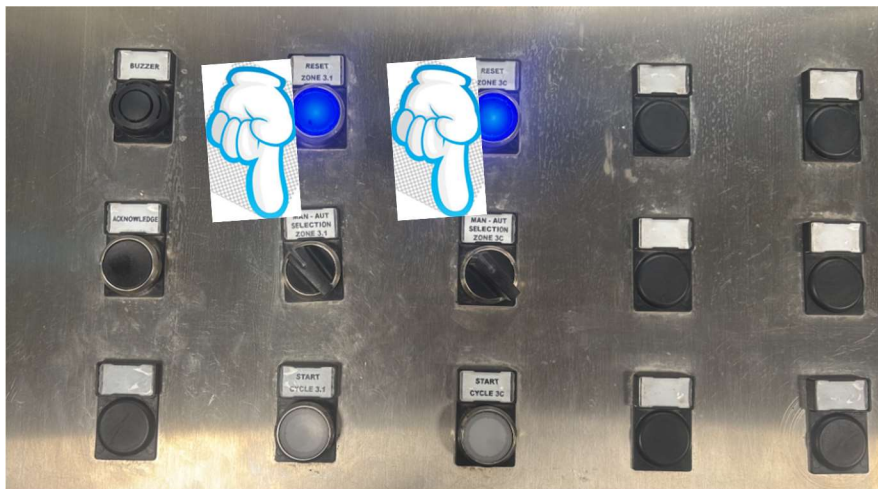
Step 1: Change Operating Mode

1. On the in-line control panel, switch **Zone 3.1** and **Zone 3C** from **AUTO** to **MANUAL**.

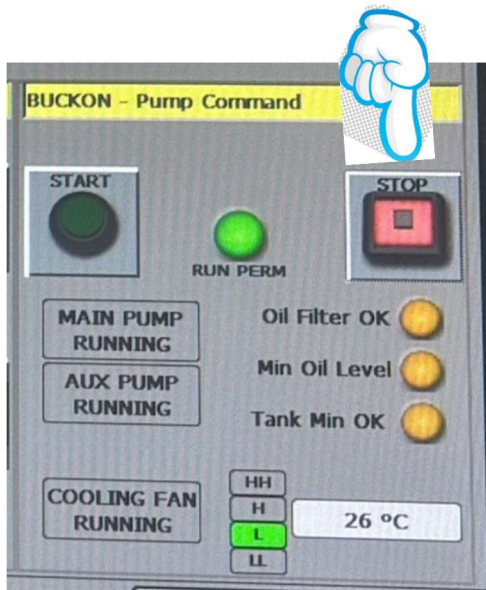
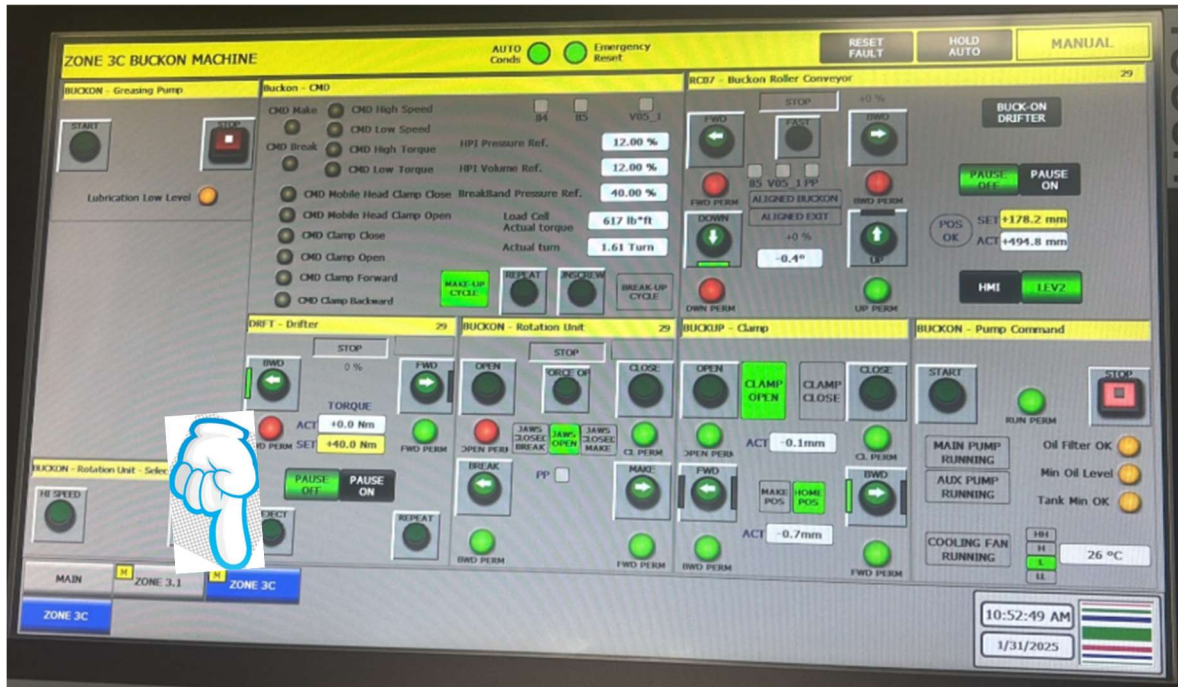
Auto Mode



Manual Mode



2. Navigate to **Section 3C** on the **Touch Panel** and stop the **Buck-on Pump**.

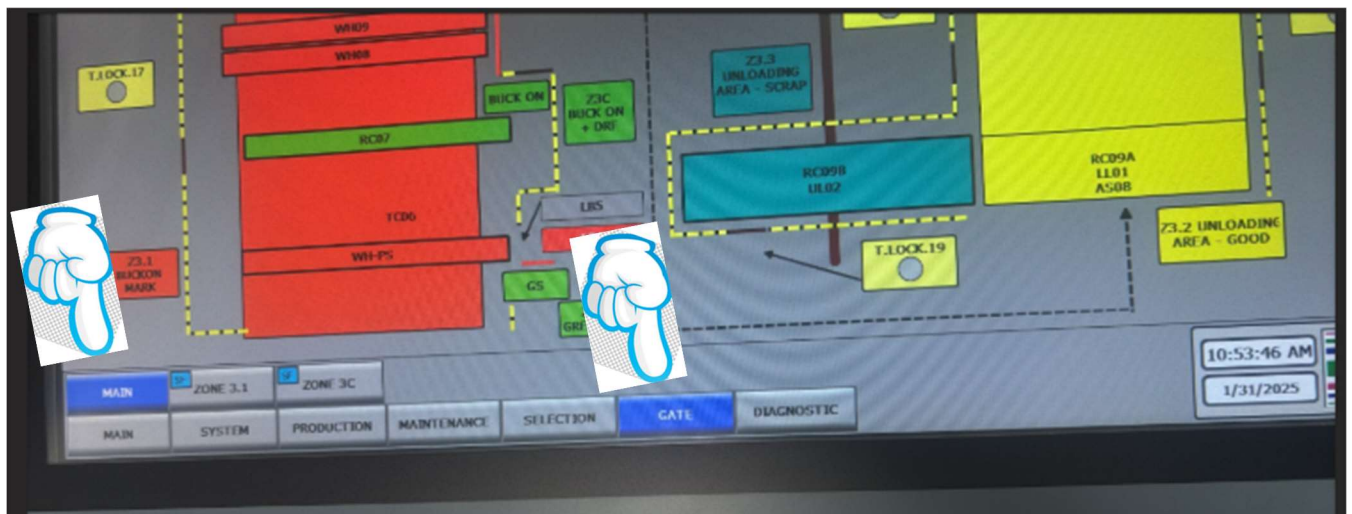


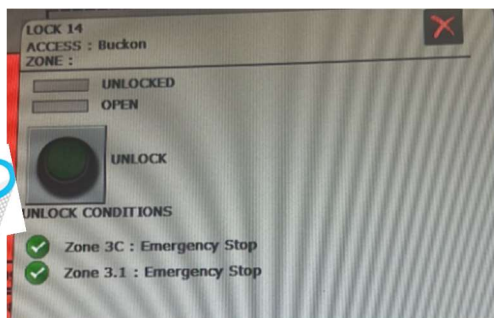
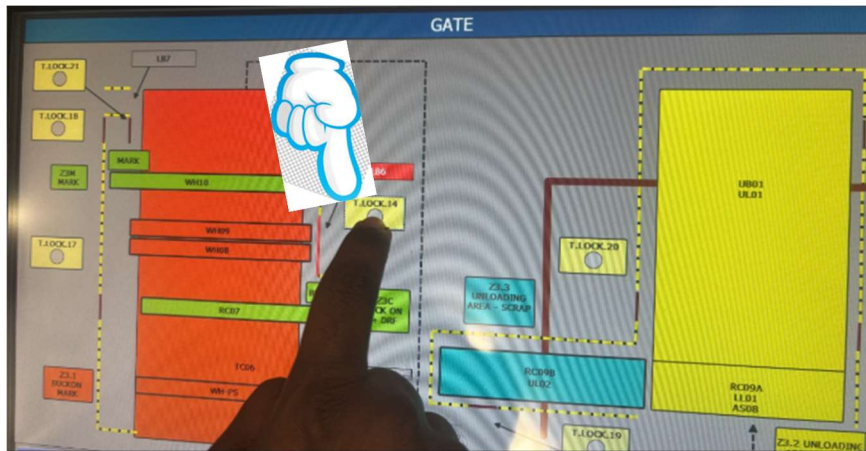
Step 2: Safety Precautions

1. Engage the **black Emergency Stop (E-Stop)** for **Zone 3.1** and **Zone 3C**.

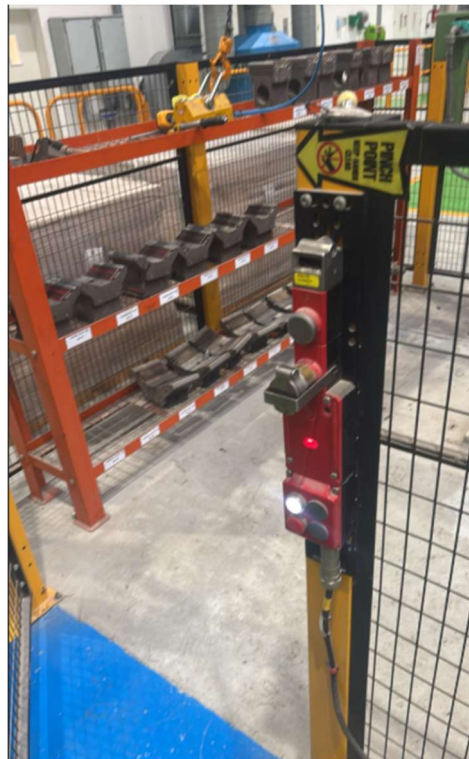


2. On the **Touchscreen Panel**, go to **Main → Gate** and unlock the door by pressing **T.LOCK.14**.





3. Manually open the gate.



Step 3: Removal of Jaws

1. Loosen the screws of the **jaw lockers** using a **6mm Allen key**.



2. Once loosened, completely remove the bolts and slide out the jaw from the base.
3. Repeat the process for all positions in both the **Rotary Head** and **Back Tong**.
4. **Safety Precautions:**

-  Risk of being hit, pinched or cut if a spanner comes off.
-  Use the hand tools properly and use them in good condition.
-  The screws are stuck and cannot be loosened
-  Pinching. -Don't loose sight of where you're putting both hands
-  Machine blocked by Tag out card.

- **Step 4: Cleaning**

- Use support of helper to transfer Jaws



- Clean the Bases of all Jaws.

5.  Risk of damage and eye injuries.

6.  During cleaning, use the safety glasses is a specific PPE.

7.  Risk of hand injuries

8.  Use safety gloves and clean by cotton rags. Watch out for the metal sharp edges.

9.  Machine blocked by Tag out card.

Step 5: Mechanical Adjustments

Centricity Adjustment of the Rotary Head

- Before starting a job, adjust the **rotary head centricity** based on the pipe diameter.
- Set the **clamping path** as per **Table 4.3** before installing the clamping jaws.
- Ensure the **rotary head is fully open (0° crank angle)** before adjusting.
- Insert a **12mm Allen wrench** into the jaw guide and adjust by turning clockwise or counterclockwise.
- Verify the settings using the **scale at each jaw guide**.

Table 4.3

Nominal Size Pipe Diameter (inch)	Coupling Outside Diameter	Clamping Path		Jaw No.		Clamping Angle Rotary Head (Coupling inserted)
		mm	at Clamping Angle	Rotary Head	Backup Tong	
9-5/8				2	2	
10-3/4				2	2	
11-3/4				1	1	
13-3/8				1	1	

Clamping Path Adjustment Table for 9-5/8 to 13 3/8-inch Diameter with Standard Jaws (continued)

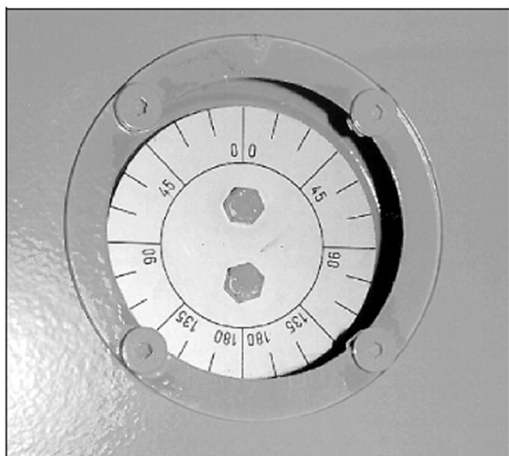
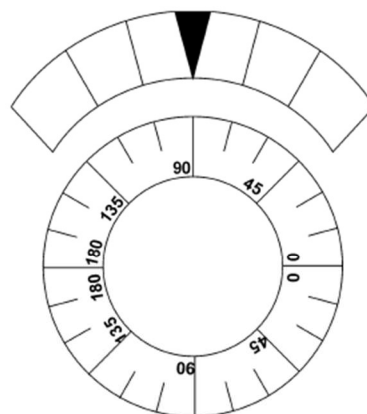
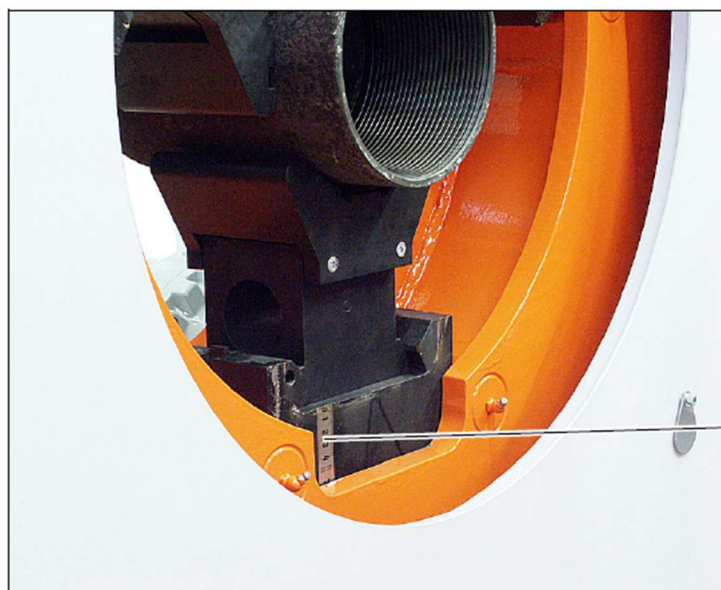


Fig. 4.8 Dial to control the Clamping Angle



Clamping angle = 90°

Fig. 4.9 Example how to read the Dial



Note: Remove the jaws and insert a 12 mm Allen wrench into the hole in the middle of the jaw guide to adjust the centricity.

Scale in Millimeter

Fig. 4.10 Adjustment of the Clamping Angle

Step 6: Installation of New Jaws

1. With the assistance of a helper, place each jaw into the **jaw base**.
2. Insert and tighten all bolts securely.
3. Repeat the process for all jaws.
4. Using a **spanner**, ensure all bolts are properly tightened.
5. Adjust the **jaw height** as per the reference manual.





Step 7: Clamping Angle Adjustments

- **For small diameters:** The **clamping angle** should be $135^{\circ} \pm 15^{\circ}$ or $45^{\circ} \pm 15^{\circ}$.
- **For all other sizes:** Adjust the angle to approximately 90° to minimize die penetration.
- If **coupling slippage** occurs, increase the angle to 110° or 135° for greater clamping force.

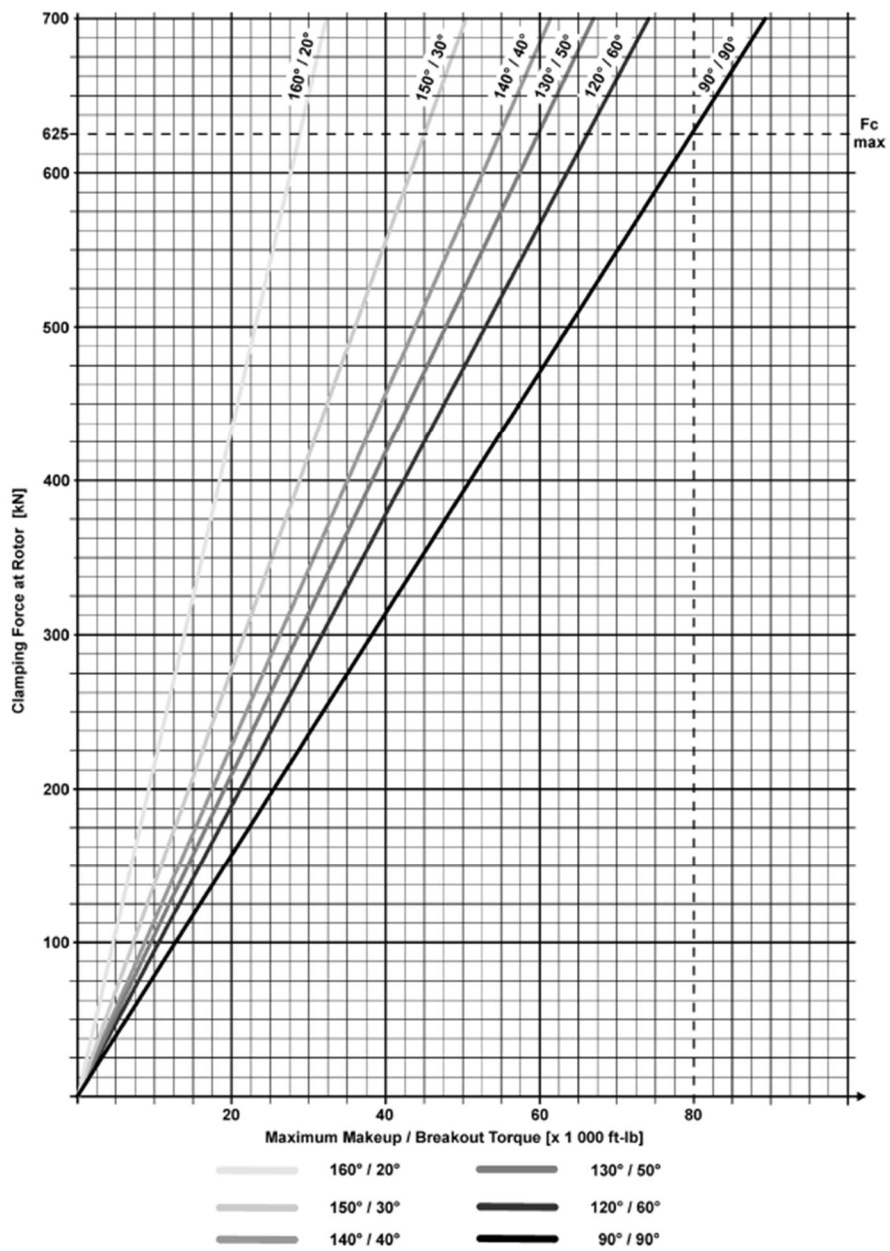


Fig. 3.17 Clamping Force Diagram

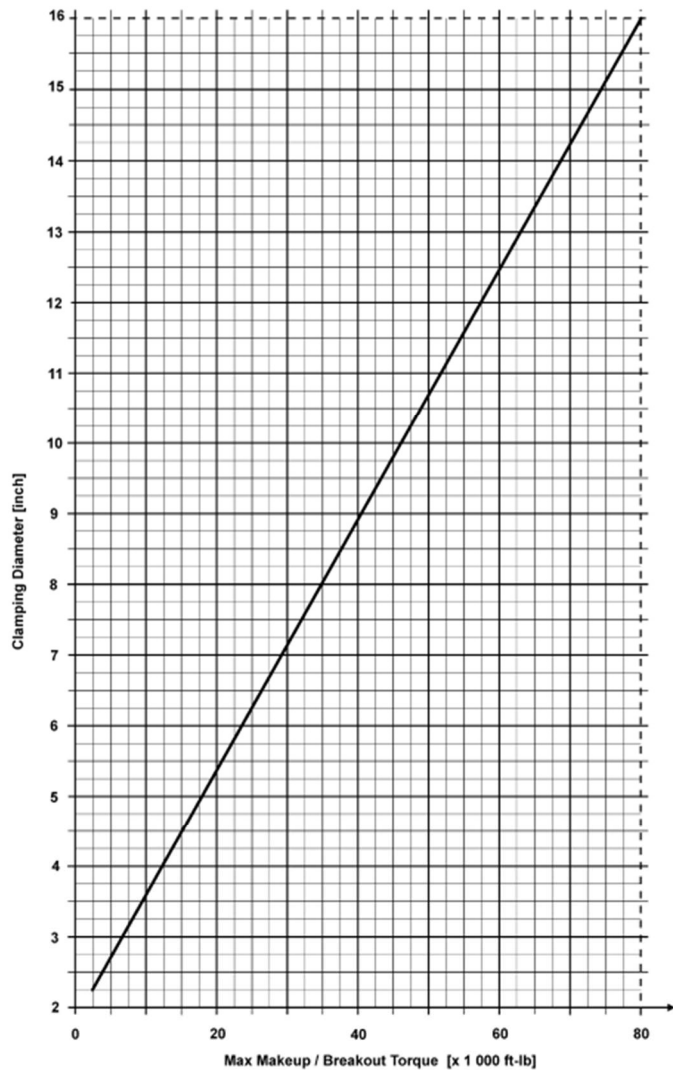


Fig. 3.18 Max. Torque Diagram depending on Clamping Diameter



The max. torque values depend on various factors such as the ability to transmit the torque through the die/pipe surface interface. For certain occasions and sizes it is possible to achieve higher torques but this might reduce the lifetime of the machine.

If the coupling slips when torque is applied, the clamping angle must be adjusted to a higher value (e.g. from 90° to 110° or even to 135°) to get a higher clamping force.

-  Risk of being pinched by Allen Key
-  Use the hand tools properly and make sure they are in good condition

-  Do not set the clamping angle of the rotary head higher than necessary to avoid deep die markings on the pipe surface.
-  Do not adjust the clamping path greater than 30 mm for torques lower than 10000 ft-lb and 25 mm for torques higher than 10000 ft-lb.
-  Pinching. -Don't loose sight of where you're putting both hands
-  Trip Hazard-Watch your step while walking, never put the jaws on the floor
-  Machine blocked by Tag out card.

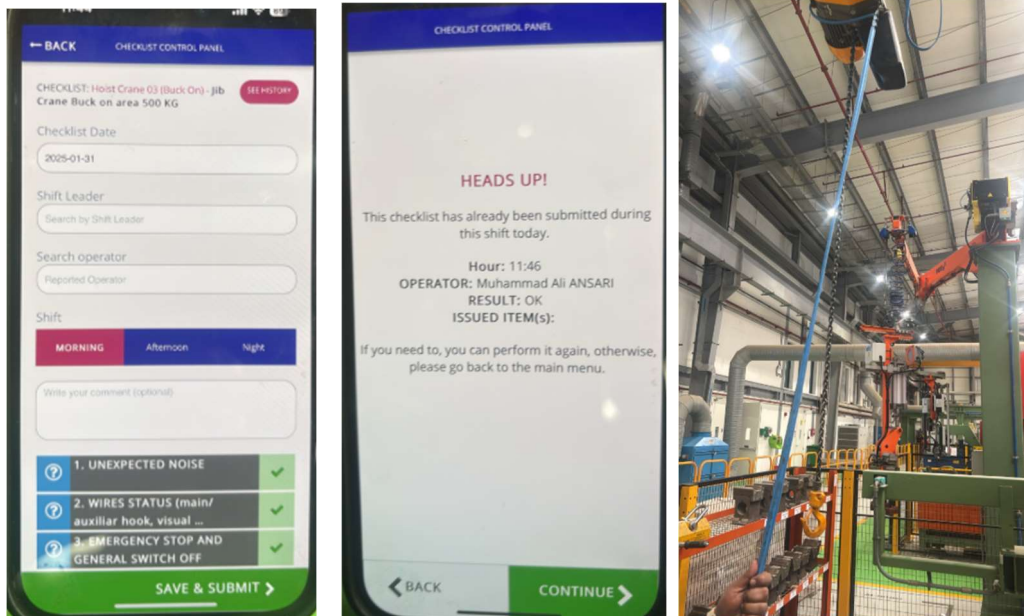
Step 8: Drift Change

1. Remove the Old Drift:

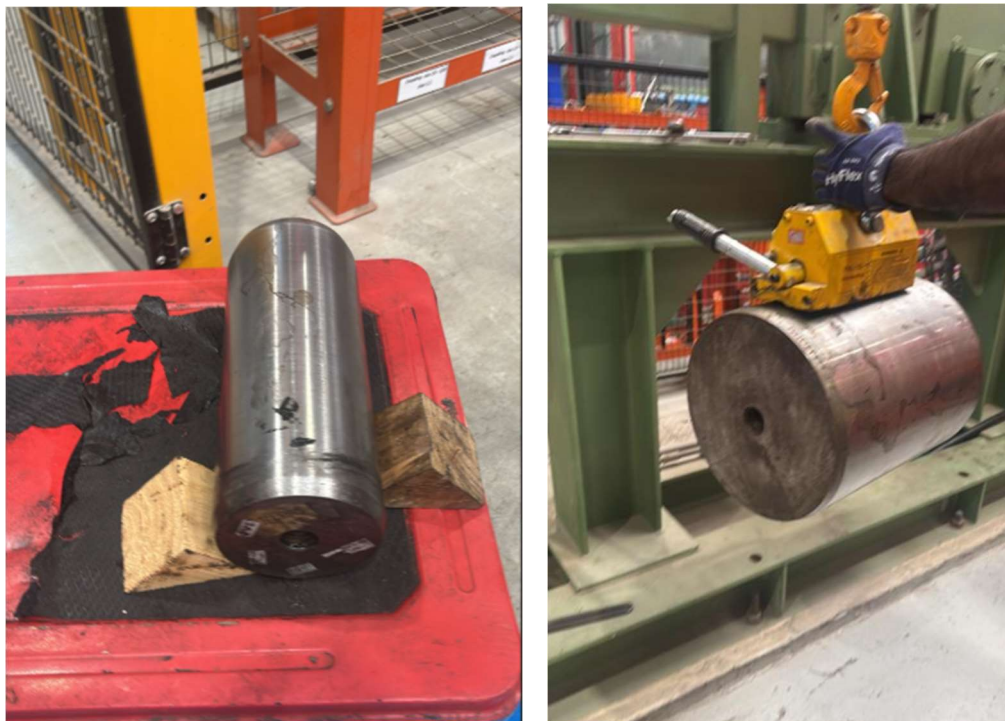
- Unscrew the bolt securing the drift.



- Use a crane (ensure operator certification and inspection before use).



- Attach the **magnetic yoke** to the drift and conduct a **validation lift (30cm above the drift table)** to confirm a secure hold.

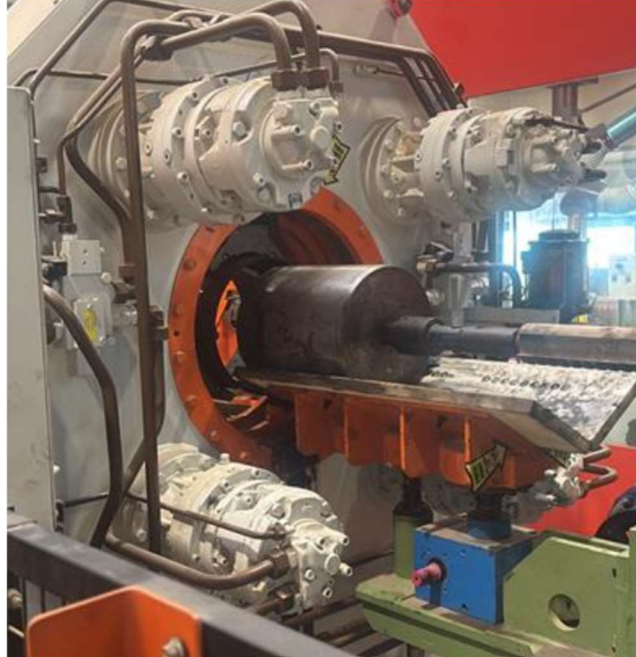


- Guide the crane to move the drift out of the machine.
- Release the **magnetic yoke pin** once the drift is properly supported.

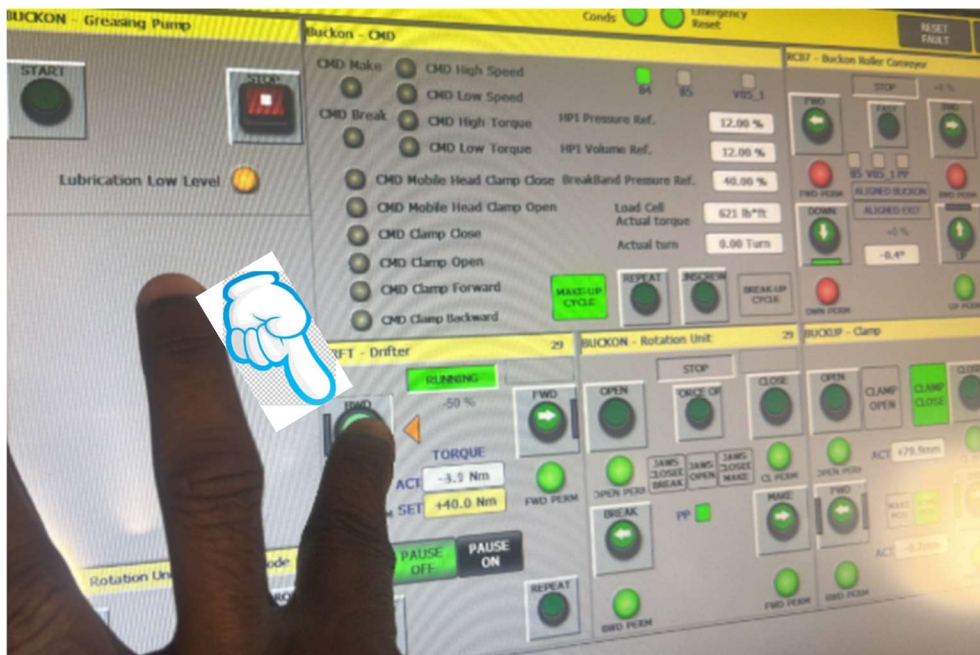


2. Install the New Drift:

- Secure the **new drift shaft** by tightening it clockwise.
- Adjust the **drift table height** to align with the **pipe centerline**.



- On the **Touch Panel**, move the drift forward to the coupling to verify alignment.





Step 9: Final Setup and Activation

1. Close the gate.
2. Switch the **manual mode** back to **AUTO mode** on the control panel.